

# Science

Nature Walks & Scouting

General Science

Natural History

Nature Notebook





## About the Course

In level 3, learners are developing familiarity and friendship with Creation through a broader scope of Things, more challenging text, and more abstract concepts.

This course includes the following topic(s): General Science: Grade 3, Natural History: Grade 3, Nature Notebook: Grade 3, Nature Walks & Scouting: Grades 1-8

**About Nature Walks & Scouting: Grades 1-8**

Students spend time once each week on a longer excursion. Prompts can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

**About General Science: Grade 3**

Students explore more complex ideas, such as ecosystems, weather and seasonal changes, water, food chains, life cycles, adaptation, and migration. Coordinating afternoon activities are provided in Outdoor Work.

**About Natural History: Grade 3**

Students learn methodically about the variety of creatures around them, including insects, birds, and reptiles. Coordinating afternoon activities are provided in Outdoor Work.

**About Nature Notebook: Grade 3**

Students spend time at least once each week recording their observations. Prompts that coordinate with curricular content can be found in Outdoor Work, if desired. This digital resource is provided with all science lessons.

**Science: Grade 3**

**The Big Picture:**

To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Form 1 learners in grades 1-3 are at the beginning of this relationship: developing familiarity and friendship with Creation. Their sense of community is primarily focused on their immediate environment at this stage, but they are curious about others. The familiarity with foundational knowledge grown and nurtured in Form 1 prepares them for greater curiosity and personal interest in Form 2.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Literature Grades 1-3).



## Placement & Combining Tips

**Science: Grade 3**

This level 3 course is for third-grade students or possibly hungry second-graders or fourth-graders taking their time. If combining grades is desirable, it is recommended that the combined grades begin at the appropriate level and move forward together.

**Nature Walks & Scouting: Grades 1-8**

Learners may be combined and share prompts or follow the plan of their local scouting troop or natural history club. Support accessibility and sensory development as appropriate.

Nature Notebook: Grade 3

Learners may be combined and share prompts or follow their own interest. Support through varied media, scribing observations, or notebooking in tandem.



## Scheduling

| GRADE | SCHEDULE INFO.                                             | BOOKS                                                          |
|-------|------------------------------------------------------------|----------------------------------------------------------------|
| 1-8   | Nature Walks & Scouting: Grades 1-8<br>1 time/week 30 min+ |                                                                |
| 3     | General Science: Grade 3<br>1 time/week 15 min             | Buzztail and Leaper<br>Next Time You See a Cloud<br>Going Home |
| 3     | Natural History: Grade 3<br>1 time/week 10 min             | Christian Liberty Nature Reader Book 4                         |
| 3     | Nature Notebook: Grade 3<br>1+ time/week 10 min+           |                                                                |

### Sample Weekly View

| Day 1                                                           | Day 2                    | Day 3                    | Day 4 | Day 5 |
|-----------------------------------------------------------------|--------------------------|--------------------------|-------|-------|
| <b>Science: Grade 3</b>                                         |                          |                          |       |       |
| General Science: Grade 3<br>Nature Walks & Scouting: Grades 1-8 | Nature Notebook: Grade 3 | Natural History: Grade 3 |       |       |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 3

- ☐ Obtain any supplies indicated on the science or grade-level supply lists.
- ☐ Download any apps and shortcut any desired links.

Nature Lore:

- ☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
- ☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

### Special Topics & Field Trips

Science: Grade 3

- ☐ Learn a few of your local species on these special topics using the internet, field guides, or your local nature preserve, conservancy, or state/national park.
- ☐ Note whether there are particular locations nearby for the student to easily notice the special topics on walks or during afternoon occupations.

#### General Science: Grade 3

- ☐ Term 1: fruits, trips in evenings, and a pet store with snakes, a herp (reptile/amphibian) conservancy, or a natural history museum; Week 11 suggests looking for snake habitat. If you are anxious about snakes, consider a program at a local nature preserve or state/national park instead.
- ☐ Term 2: clouds, trips in evenings, an owl/raptor conservancy/wildlife rehabilitator or natural history museum, and an aquarium/pet store/fish market at a grocery store, or a natural history museum
- ☐ Term 3: fish, birds, tree buds, trips to a pond, and a natural history museum

#### Natural History: Grade 3

- ☐ Term 1: insects and birds
- ☐ Term 2: birds and trees
- ☐ Term 3: turtles, fish, birds, tree buds, includes special excursions to a pond and a natural history museum

### Term Prep & Teacher Tips

#### Science: Grade 3

Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term:

#### General Science: Grade 3

- ☐ Term 1: snakes p.193-203, ladybugs p.364-366
- ☐ Term 2: clouds/water forms p.808-814, screech owls p.100-103
- ☐ Term 3: migrating birds p.35-38, bird beaks p.39-40, and/or choose 1-2 local birds from p.50-143

☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:

- means to make a small collection (small box or selection of boxes, photo album, etc)

#### Natural History: Grade 3

- ☐ Term 1: moths p.310-328
- ☐ Term 2: crows p.124-127
- ☐ Term 3: turtles p.204-210

### Reminders

#### General Science: Grade 3

- ☐ Students will raise ladybugs at the start of Term 1, so make a note in your calendar to order them at the appropriate time and be flexible with the schedule to accommodate their development and release according to the instructions that come with them.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Science: Grade 3

#### General Science: Grade 3



Buzztail and Leaper



Next Time You See a Cloud



Going Home

## Natural History: Grade 3



Christian Liberty Nature Reader Book 4



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

### Nature Walks & Scouting: Grades 1-8



Wild Bird Seed



Bird Feeder



Hand Lens



Small Collection Containers



Small Basin



Dip Net



Student Microscope

### General Science: Grade 3



Household Items - General Science: Grade 3



Ladybug Land with Voucher to Obtain Larva



Large Owl Pellet Kit



Small Collection Containers



## Quick Links

### Science: Grade 3

∞ [Outdoor Work](#)

∞ [Seek app from iNaturalist](#)

∞ [SkyView Lite for Android](#)

∞ [SkyView Lite for iOS](#)

[Click THIS text or scan the QR code for links.](#)



# Science: Grade 3

## How To Teach



### Introduce

- Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap. The lesson plans often help with this.
- Some learners may benefit from using pictures or looking back briefly.
- Allow them time to share any concerns and come alongside, as needed.



### Complete

- Read, or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their nature journal or some other notebook/paper available in case they need to draw or diagram during the lesson.
- When reading to pre-readers, point to the words as you read, even if it seems like they aren't paying attention.
- If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, or beyond.



### Narrate

- Process the ideas of the lesson by retelling the events in sequence, describing a scene or setting, explaining a concept, etc.
- Learners may use words, pictures, Legos, dolls, etc., to process and convey ideas in the manner most natural to them.
- Teachers may take turns to model.



### Discuss

- Consider together any thoughts, confusion, or concerns about the passage.
- Questions/topics for further discussion are often provided in the lesson plans to help. There are no right or wrong answers to these. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.



### Connect

- Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner interest.
- These can also be viewed as alternative ways to engage.

## Term 1

### WEEK 1

#### 15m General Science: Grade 3 - Lesson 1

*Noticing Clouds*

Materials: Next Time You See a Cloud

PREP: Read Teacher Tip

##### → INTRO

Look at the picture on the cover. Have you watched clouds before? What do you think it would feel like to touch a cloud?

##### → READ, NARRATE, & DISCUSS

p.7-18 "Next time you" - "about to change."

- We call the living Things the biome. The living and non-living parts of the environment together are called the ecosystem. Water is one of those non-living parts of the ecosystem. Can you think of some ways that water affects the living Things?

##### ★ TEACHER TIP

Let students enjoy the photographs in this picture book and take time to talk about them.

##### ★ TEACHER NOTE

If you haven't ordered your larvae yet - don't forget!

##### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 1

#### 10m Natural History: Grade 3 - Lesson 1

*The Bumblebee*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

##### → INTRO

Look at the picture on p.87. What do you notice, or what do you already know about the bumblebee? In our reading today, the author states that bumblebees are the only bee native to North America, but that's not quite true. They are the only social bee, but there are some 3600 other native North American bees! Let's read and try to learn something new about this one today.

##### → READ, NARRATE, & DISCUSS

p.86-90 "Sometimes" - "worker or queen."

- You may have read about or noticed before that some creatures depend on each other, like a butterfly and their host plants. Based on what you read, what other plants or animals could you say the bumblebee depends on? What plants or animals depend on the bumblebee?

##### → SUPPLEMENTAL

∞ Video Link: Bumblebee in Slow Motion (1:40)

##### ★ TEACHER NOTE

This author states that bumblebees are the only native bee in North America, but this is false. Teachers can strike the third sentence (if reading aloud) or advise students as noted in the introduction.

##### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 2

#### 15m General Science: Grade 3 - Lesson 2

*Observing Ladybug Larvae Activity*

Materials: Nature Journal, Ladybug larvae

PREP: Read Teacher Tip

##### → INTRO

Spend about 10 minutes observing your ladybug larvae in their habitat today and plan to check on them every day to watch them change. What parts are you able to notice? Are there visible body segments? Are there legs? If so, how many? Are there eyes? Antennae? What are the larvae doing?

##### → OBSERVE

##### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- A habitat must provide the food, water, and shelter that a living Thing needs, but it is very different from a real ecosystem. How does your habitat provide what the ladybug larvae need? How is it different from the ladybug's natural ecosystem?

##### ★ TEACHER TIP

If students seem anxious or unsure about their nature journal, teachers can encourage them by modeling, showing them many different examples, journaling in tandem, or by serving as a scribe.

##### ★ TEACHER TIP

Raising ladybugs provides many opportunities for object lessons on several stages of the insect's life over the next two weeks or so. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.

### WEEK 2

#### 10m Natural History: Grade 3 - Lesson 2

*The Ladybug*

Materials: Christian Liberty Nature Reader 4

##### → INTRO

Recall what we read about the bumblebee. Look at the picture on p.91. What do you notice? What insect family do you think the ladybug belongs to?

##### → READ, NARRATE, & DISCUSS

p.91-94 "Although there" - "outdoors again."

- Who in this chapter would you say is dependent on whom?

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 1

### → SUPPLEMENTAL

∞ Video Link: Ladybugs in Slow Motion (2:11)

### WEEK 3

#### 15m General Science: Grade 3 - Lesson 3

*Observing Larvae or Pupae Activity*

Materials: Nature Journal, Ladybug larvae

PREP: Read Teacher Tip

### → INTRO

Spend about 10 minutes observing your larvae or pupae today and plan to check on them every day to watch them change. What do you notice today?

### → OBSERVE

### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

### ★ TEACHER TIP

The time that it takes for the ladybugs to emerge will vary slightly, so enjoy them while you have them and release them based on the instructions that came with them.

### WEEK 3

#### 10m Natural History: Grade 3 - Lesson 3

*Aphids*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

### → INTRO

Tell me about the ladybug and why everyone likes them so much. Today we read about those aphids.

### → READ, NARRATE, & DISCUSS

p.95-98 "Aphids are" - "by eating aphids."

• The author describes the problem of aphids harming plants, but everything has a purpose in Creation. Can you think of some reason to appreciate the aphid?

### → SUPPLEMENTAL

∞ Video Link: Aphids Close Up (0:44)

### ★ TEACHER TIP

It can sometimes be unnerving for students to not get an immediate answer to a question, so encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4

#### 15m General Science: Grade 3 - Lesson 4

*Observing Ladybugs and Watching the Clouds Go By Activity*

Materials: Nature Journal, Ladybugs, if not yet released

### → INTRO

Spend about 10 minutes observing your ladybugs today, if you have not yet released them. What parts are you able to notice on the adult beetle? How does it differ from the larva? What colors and markings do you notice? How does it use its wings? If you have released them already (or as you watch them fly away), notice the clouds in the sky or their absence if it is a clear day. What colors and shapes do you notice in the sky? Are the clouds moving or changing? Do you think they are cirrus, stratus, or cumulus? Or maybe some combination?

### → OBSERVE

### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

• Tell about what you recorded.

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 4

#### 10m Natural History: Grade 3 - Lesson 4

*The Lacewing*

Materials: Christian Liberty Nature Reader 4

### → INTRO

Remind me what we read about last time. Look at the picture on p.99. You may have read a bit about the lacewing before, but have you ever seen one? Can you see why it is called the lacewing?

### → READ, NARRATE, & DISCUSS

p.99-101 "The lacewing" - "destroy aphids."

• Compare the lifecycle of the lacewing to that of the butterfly or beetle.

### → SUPPLEMENTAL

∞ Video Link: Green Lacewing Close Up (0:33)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 1

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| WEEK 5 | <p><b>15m General Science: Grade 3 - Lesson 5</b><br/><i>Noticing Clouds</i></p> <p><b>Materials:</b> Next Time You See a Cloud</p> <p><b>→ INTRO</b><br/>What kinds of clouds have you noticed since we last read? Let's read some more about clouds today.</p> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>p.19-29 "As more and more" - "that remarkable?"</p> <ul style="list-style-type: none"> <li>Where does the rain that condenses from the clouds go? Find a major river on your globe that is near the student's home/school and trace its path to the ocean. If you were a drop of water in that river, what kinds of plants and animals might you pass on your way? What if one of them drank you? If you made it the whole way to the ocean, then what?</li> <li>Diagram the water cycle showing how that water droplet gets back up into the clouds.</li> </ul>                                                                                                                                      | <p><b>• NATURE NOTEBOOK</b><br/>Prompt for General Science in Outdoor Work Quick Link.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| WEEK 5 | <p><b>10m Natural History: Grade 3 - Lesson 5</b><br/><i>The Mosquito</i></p> <p><b>Materials:</b> Christian Liberty Nature Reader 4</p> <p>PREP: Read Teacher Tip</p> <p><b>→ INTRO</b><br/>Can you recall 3 insects that you read about so far this term? Today, we are going to read about the mosquito. Look at the picture on p.102. What do you notice? What about the mosquito's life cycle is similar to or different from the other insects you know?</p> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>p.102-106 "If one looks" - "in the spring."</p> <ul style="list-style-type: none"> <li>The author mentions that the development of the mosquito depends on the temperature, and you can see that they need water during the first 3 stages of their life cycle. Based on that information, where do you suppose mosquitoes would be more or less of a problem? You can use a map or globe for help, if you like.</li> </ul> <p><b>→ SUPPLEMENTAL</b><br/>∞ Video Link: Mosquito Close Up (0:24)</p> | <p><b>★ TEACHER TIP</b><br/>If students have difficulty considering the discussion question because they have little awareness of climate in different places, that's okay! Show them on the globe that because the sun's light hits the equator directly, those places are hotter, whereas the poles are colder. Just let them hypothesize about where there might be more mosquitoes and leave it at that!</p> <p><b>• OUTDOOR WORK</b><br/>Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p> |
| WEEK 6 | <p><b>15m General Science: Grade 3 - Lesson 6</b><br/><i>Observing Clouds Activity</i></p> <p><b>Materials:</b> Nature Journal</p> <p><b>→ INTRO</b><br/>Spend about 10 minutes observing clouds. What colors and shapes do you notice in the sky? Are the clouds moving or changing? Do you think they are cirrus, stratus, or cumulus? Or maybe some combination? Does their appearance change from one part of the sky to another?</p> <p><b>→ OBSERVE</b></p> <p><b>→ NARRATE &amp; DISCUSS</b><br/>Use a nature journal to record what the student noticed.</p> <ul style="list-style-type: none"> <li>Tell about what you recorded.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                | <p><b>• NATURE NOTEBOOK</b><br/>Prompt for General Science in Outdoor Work Quick Link.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| WEEK 6 | <p><b>10m Natural History: Grade 3 - Lesson 6</b><br/><i>The Dragonfly and Damselfly</i></p> <p><b>Materials:</b> Christian Liberty Nature Reader 4</p> <p>PREP: Read Teacher Tip</p> <p><b>→ INTRO</b><br/>Tell me what you remember about the life cycle of the mosquito. Look at the picture on p.107. What do you notice or recall about the dragonfly?</p> <p><b>→ READ, NARRATE, &amp; DISCUSS</b><br/>p.107-112 "Let us watch" - "do not hurt it."</p> <ul style="list-style-type: none"> <li>How do you think the weather and seasons affect the life cycle of the dragonfly?</li> </ul> <p><b>→ SUPPLEMENTAL</b><br/>∞ Video Link: Dragonfly or Damselfly? (0:55)</p>                                                                                                                                                                                                                                                                                                                                      | <p><b>★ TEACHER TIP</b><br/>Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).</p> <p><b>• OUTDOOR WORK</b><br/>Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p>           |

## Term 1

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| WEEK 7 | <b>15m General Science: Grade 3 - Lesson 7</b><br><i>Caterpillars and Pupae Activity</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|        | <div>Materials: Nature Journal</div> <p>→ <b>INTRO</b><br/>Spend about 10 minutes looking at and/or observing caterpillars and pupae. Some good places to look might be underneath the leaves of plants, around the base of plants, and on twigs or branches. If you find one, notice its colors and patterns, size, and any nearby plants. Collect as many clues as you can about who this might be.</p> <p>→ <b>OBSERVE</b></p> <p>→ <b>NARRATE &amp; DISCUSS</b><br/>Use a nature journal to record what the student noticed.</p> <ul style="list-style-type: none"> <li>What are some ways that you could find out who this is?</li> </ul>                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| WEEK 7 | <b>10m Natural History: Grade 3 - Lesson 7</b><br><i>The Tomato Hornworm</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>★ <b>TEACHER TIP</b><br/>If students are interested in the idea of the food chain, teachers can add that many birds like to eat hornworms, too!</p> <p>★ <b>TEACHER TIP</b><br/>If students need support drawing or diagramming, the teacher may do the activity with them in tandem or act as their scribe.</p> <p>• <b>OUTDOOR WORK</b><br/>Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p> |
|        | <div>Materials: Christian Liberty Nature Reader 4</div> <p>PREP: Read Teacher Tip</p> <p>→ <b>INTRO</b><br/>Tell me about 3 of the insects we have read about so far. Look at the picture on p.114. What stands out to you about this caterpillar?</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.113-116 "If you have" - "full-grown."</p> <ul style="list-style-type: none"> <li>How are the tomato plant, the hornworm, and the parasitic wasp connected? Draw a diagram, called a food chain, if helpful. Who eats whom? What if something in the ecosystem changed and all of the parasitic wasps died? How do you think that would affect the other members of the food chain?</li> </ul> <p>→ <b>SUPPLEMENTAL</b><br/>∞ Video Link: Hornworm Up Close (0:44)</p> |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| WEEK 8 | <b>15m General Science: Grade 3 - Lesson 8</b><br><i>A Creature in Context</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>★ <b>TEACHER NOTE</b><br/>Page 13 doesn't actually have the page number, so you may need to indicate it before the reading.</p>                                                                                                                                                                                                                                                                                            |
|        | <div>Materials: Buzztail and Leaper</div> <p>→ <b>INTRO</b><br/>This is two books in one, so we are going to start with Buzztail. Have you ever come across a snake before when playing or on a walk? If so, tell about it. What did you think of the snake?</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.3-13 "The rattlesnake" - "would come along."</p> <ul style="list-style-type: none"> <li>Talk about a food chain based on what you read. You might diagram, if desired. Do you have these same animals or similar ones in your biome? What other animals would be in yours? Do you know which of these creatures are cold-blooded and which are warm-blooded? When it's too cold for cold-blooded creatures, how does that affect the food chain?</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| WEEK 8 | <b>10m Natural History: Grade 3 - Lesson 8</b><br><i>The Hornworm Changes</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>★ <b>TEACHER TIP</b><br/>Engage students in discussion to help them understand that the ecosystem is complex.</p> <p>• <b>OUTDOOR WORK</b><br/>Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p>                                                                                                                                                                                                |
|        | <div>Materials: Christian Liberty Nature Reader 4</div> <p>PREP: Read Teacher Tip</p> <p>→ <b>INTRO</b><br/>Tell me what you remember about the hornworm. Today, let's read about its metamorphosis. What do you recall about metamorphosis?</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.117-120 "When full-grown" - "tomato plants."</p> <ul style="list-style-type: none"> <li>In what ways would you say that the creatures in this chapter are good or bad?</li> </ul> <p>→ <b>SUPPLEMENTAL</b><br/>∞ Video Link: Sphinx Moth (1:04)</p>                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Term 1

### WEEK 9

#### 15m General Science: Grade 3 - Lesson 9

*Buzztail's Adaptations*

Materials: Buzztail and Leaper

##### → INTRO

Where did we leave Buzztail in the last passage? Let's find out something more about him today.

##### → READ, NARRATE, & DISCUSS

p.14-24 "Every so often" - "female snake again."

- Diagram the snake, discussing the special adaptations described in the story. Adaptations are special behaviors or body parts that fit a creature to its purpose in the ecosystem.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 9

#### 10m Natural History: Grade 3 - Lesson 9

*The Cecropia Moth*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

##### → INTRO

Do you recall how the author said that moths are different from butterflies? Look at the picture on p.122-123 (it is also the picture on the cover). What do you notice about this Cecropia moth? You can continue looking at it while we read.

##### → READ, NARRATE, & DISCUSS

p.121-125 "The cecropia" - "to the twig."

- Cecropia lives in eastern North America, where there are hardwood trees, like maple. What would happen if something in the environment changed that caused the maple trees to die?

##### → SUPPLEMENTAL

- ∞ Image Link: University of Florida: Cecropia Stages
- ∞ Video Link: Cecropia Moth (0:27)

#### ★ TEACHER TIP

Sometimes readings can seem harder to narrate when there are no events that fit together in sequence or when students are unable to create the image described in their mind. It can be helpful to draw during the reading or to provide an image, like the ones linked in this lesson, to which they can attach the ideas during the reading. They can then use the image, either from memory or by pointing to it, during their narration.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 10

#### 15m General Science: Grade 3 - Lesson 10

*Buzztail's Habitat*

Materials: Buzztail and Leaper

##### → INTRO

Recall what you heard about Buzztail in the last passage.

##### → READ, NARRATE, & DISCUSS

p.25-35 "Something else" - "to attack him."

- What have you heard so far about the snake's habitat? Does he live in just one? Who else lives in this biome with the snake?

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 10

#### 10m Natural History: Grade 3 - Lesson 10

*The Caterpillar Changes*

Materials: Christian Liberty Nature Reader 4

##### → INTRO

Tell me what you recall about the Cecropia caterpillar and moth.

##### → READ, NARRATE, & DISCUSS

p.126-128 "Soon the caterpillar" - "work is done."

- Cecropia moths emerge in the spring as the maples are growing their new leaves and flowers. How do you think the colors and patterns of this caterpillar and moth serve it?

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 11

#### 15m General Science: Grade 3 - Lesson 11

*Observing Habitat Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

##### → INTRO

Spend about 10 minutes observing the habitats near you. Are there any that would be good homes for snakes? (You are thinking about the habitat, not necessarily searching for snakes.) Remember to always use a stick to help if you want to look under something, so that your fingers don't surprise a hidden creature. If you want to lift up a rock or log, always lift the side that is away from you. If you come across a snake, keep your distance and don't corner it unless you are sure it is safe. Describe the habitat. What other living Things live there?

#### ★ TEACHER TIP

This could be a good opportunity for an object lesson with a snake, if the student comes across one that is dormant or dead. Be sure to notice if the snake is responding to changes in the weather or season.



## Term 1

### → OBSERVE

### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- Tell about what you recorded.

### WEEK 11 10m Natural History: Grade 3 - Lesson 11

*The Isabella Tiger Moth*

Materials: Christian Liberty Nature Reader 4

### → INTRO

Remind me what the hornworm and Cecropia caterpillar look like. Have you ever seen a fuzzy black and brown caterpillar? Do you know what kind of moth he becomes? Let's read about that one today.

### → READ, NARRATE, & DISCUSS

p.129-132 "When one goes" - "eat them at night."

- How do you think the colors and patterns of this caterpillar and moth serve it? Make a food chain diagram for this creature. What if changes in the environment killed the clover and dandelion? What if changes killed the bats?

### → SUPPLEMENTAL

∞ Video Link: Isabella Tiger Moth (1:53)

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 12 15m General Science: Grade 3 - Lesson 12

*Exam Week*

### → EXAMS

Answer question(s) related to course.

Questions will come from:

Next Time You See a Cloud, Buzztail and Leaper, Nature Journal

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 12 10m Natural History: Grade 3 - Lesson 12

*Exam Week*

### → EXAMS

Answer question(s) related to course.

Questions will come from:

Christian Liberty Nature Reader 4

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 2

### WEEK 1

#### 15m General Science: Grade 3 - Lesson 13

*Observing Seasonal Changes Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

##### → INTRO

Spend about 10 minutes observing two easy-to-see signs of the season: clouds and birds. What colors and shapes do you notice in the sky? What colors and patterns do you see in the birds? Are the clouds moving or changing? Do you think they are cirrus, stratus, or cumulus? How do the birds behave? Do you know these birds? What colors or markings do you notice? Do the clouds and birds seem different today than they did last term?

##### → OBSERVE

##### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- Tell about what you recorded.

##### ★ TEACHER TIP

There is no implied connection between clouds and birds; they are simply Things that can signal seasonal change. If teachers would like to do an object lesson on birds or clouds, this could be a good day. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.

### WEEK 1

#### 10m Natural History: Grade 3 - Lesson 13

*The Quail*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

##### → INTRO

Look at the picture on p.1. Have you ever seen a bird like this one before?

##### → READ, NARRATE, & DISCUSS

p.1-4 "Did you ever" - "ice and snow."

- What can you tell about the quail's habitat from the reading? What living and non-living parts of the ecosystem do you think affect the quail most?

##### → SUPPLEMENTAL

∞ Audio & Image Link: All About Birds: Bobwhite (a Quail)

##### ★ TEACHER TIP

This passage calls attention to the uniqueness of birds' songs. Listening to recordings, like those found at the link, will help students learn to recognize and find the birds in nature. These links are provided for all of our birds this term. Teachers will find related species from other parts of the continent near the bottom of each page, and are encouraged to explore the images and audio for the species near them.

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 2

#### 15m General Science: Grade 3 - Lesson 14

*How Buzztail Spends the Season*

Materials: Buzztail and Leaper

PREP: Read Teacher Tip

##### → INTRO

Do you recall where we left Buzztail?

##### → READ, NARRATE, & DISCUSS

p.36-45 "Buzztail struck" - "force something out."

- Can you tell what time of year it is in the story? Why is the snake basking in the sun? What do you think the female rattlesnake is going to 'force out'?

##### ★ TEACHER TIP

It has been several weeks since students left Buzztail, so they may need some help recalling where the story left off. It is okay to look back at the pictures on previous pages or even read the last few sentences from p.35, if they need this help.

### WEEK 2

#### 10m Natural History: Grade 3 - Lesson 14

*The Brown Thrasher*

Materials: Christian Liberty Nature Reader 4

##### → INTRO

Tell me something that you recall about the quail. Today's bird is quite a bit smaller than the quail. Look at the picture on p.7. What stands out to you about the brown thrasher?

##### → READ, NARRATE, & DISCUSS

p.6-11 "Listen to that" - "in the spring."

- Compare the brown thrasher and the quail. How are they alike and how are they different?

##### → SUPPLEMENTAL

∞ Audio & Image Link: All About Birds: Brown Thrasher

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 2

### WEEK 3

#### 15m General Science: Grade 3 - Lesson 15

*Life Cycle of a Snake*

Materials: Buzztail and Leaper, Nature Journal

##### → INTRO

What did Buzztail see on his journey? Are you ready to find out what happens to the female rattlesnake?

##### → READ, NARRATE, & DISCUSS

p.46-55 "At last a" - "swollen and discolored."

##### → DIAGRAM

Begin to diagram the life cycle of the rattlesnake (you will continue over the next few weeks). Compare its lifecycle to that of an insect and a bird. Many reptiles, insects, and birds are oviparous - they are egg (ovo-) producing (parous). Mammals, like cats, skunks, and bats, are viviparous - they are live (vivi-) producing (parous). Some creatures are ovoviviparous - they are a little of both.

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 3

#### 10m Natural History: Grade 3 - Lesson 15

*The Cardinal*

Materials: Christian Liberty Nature Reader 4

##### → INTRO

Remind me who we have read about so far. Look at the picture on p.12. What do you notice about this bird?

##### → READ, NARRATE, & DISCUSS

p.12-14 "Here is one" - "enemies of birds."

- How is the cardinal different from the quail and thrasher? How do you think cardinals' colors and patterns serve them? Do you think the cardinal is the "most interesting and attractive" bird? Why or why not?

##### → SUPPLEMENTAL

∞ Audio & Image Link: All About Birds: Cardinal

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4

#### 15m General Science: Grade 3 - Lesson 16

*Life Cycle of a Snake*

Materials: Buzztail and Leaper

PREP: Read Teacher Tip

##### → INTRO

What happened in the last passage? What do you think is going to happen next? Let's finish Buzztail's story today. The author uses the word 'hibernation' to describe what happens to the snakes over winter, but snakes don't actually hibernate like mammals because they are cold-blooded. Cold-blooded snakes brumate (i.e. become extremely slow) rather than sleep.

##### → READ, NARRATE, & DISCUSS

p.56-64 "Quickly he" - "whatever came along."

##### → DIAGRAM

Continue your lifecycle diagram, adding important parts of Buzztail's ecosystem at each stage of his lifecycle. You could show other plants and animals, special body adaptations, water, season, and/or habitat - whatever seems important to you. If you don't have enough time to finish this week, you can continue this next week.

#### ★ TEACHER TIP

If students need help understanding the difference between hibernation and brumation, it can help to imagine brumation like the change that happens when syrup is cold versus warm, whereas hibernation is more like a deep sleep.

#### ★ TEACHER TIP

Note the safety tip at the bottom of p.57.

### WEEK 4

#### 10m Natural History: Grade 3 - Lesson 16

*The Purple Martin*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

##### → INTRO

Tell me something that you remember about the cardinal. Look at the picture on p.16-17. What stands out to you?

##### → READ, NARRATE, & DISCUSS

p.16-19 "Did you ever" - "colony birdhouses."

- The author says that purple martins migrate in the fall so they can have plenty of food during winter. Explain what this means.

#### ★ TEACHER TIP

Help students understand that seasonal changes in the insect population force purple martins to migrate because of how they are designed to eat.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## Term 2

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                         |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | <p>→ <b>SUPPLEMENTAL</b><br/>∞ Audio &amp; Image Link: All About Birds: Purple Martin</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                         |
| WEEK 5 | <p><b>15m General Science: Grade 3 - Lesson 17</b><br/><i>Illustrating a Snake's Life Activity</i></p> <p><b>Materials:</b> Nature Journal, Buzztail and Leaper</p> <p>PREP: Read Teacher Tip</p> <p>→ <b>INTRO</b><br/>Recall what we know about Buzztail's lifecycle.</p> <p>→ <b>ACTIVITY</b><br/>Spend about 10 minutes finishing your lifecycle diagram.</p> <p>→ <b>NARRATE &amp; DISCUSS</b><br/>• Explain what you recorded.</p>                                                                                                                                                                                                                                                                                                                          | <p>★ <b>TEACHER TIP</b><br/>If students finished this activity last week, have them draw a picture or comic showing the difference between hibernation and brumation. Alternatively, they can act out a scene with puppets or dolls.</p>                                                                                                                                |
| WEEK 5 | <p><b>10m Natural History: Grade 3 - Lesson 17</b><br/><i>The Kingbird</i></p> <p><b>Materials:</b> Christian Liberty Nature Reader 4</p> <p>PREP: Read Teacher Tip</p> <p>→ <b>INTRO</b><br/>Remind me what we read about last time. Today, we have another bird that eats on the wing. Look at the picture on p.20. What do you notice about this bird?</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.20-22 "Have you ever" - "in the spring."</p> <p>• The author states that we should protect the kingbird because 'it is so brave and so helpful to man.' What are other reasons you can think of why we should protect any creature?</p> <p>→ <b>SUPPLEMENTAL</b><br/>∞ Audio &amp; Image Link: All About Birds: Western Kingbird</p>                 | <p>★ <b>TEACHER TIP</b><br/>Help students to understand that we should protect all of God's creatures, whether we understand their purpose or not.</p> <p>• <b>OUTDOOR WORK</b><br/>Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p>                                                                                                        |
| WEEK 6 | <p><b>15m General Science: Grade 3 - Lesson 18</b><br/><i>Observing Seasonal Changes Activity</i></p> <p><b>Materials:</b> Nature Journal</p> <p>→ <b>INTRO</b><br/>Spend about 10 minutes observing two easy-to-see signs of the season: clouds and birds. What colors and shapes do you notice in the sky? What colors and patterns do you see in the birds? Are the clouds moving or changing? Do you think they are cirrus, stratus, or cumulus? How do the birds behave? Do you know these birds? Do the clouds and birds seem different today than they did last term?</p> <p>→ <b>OBSERVE</b></p> <p>→ <b>NARRATE &amp; DISCUSS</b><br/>Use a nature journal to record what the student noticed.</p> <p>• Tell about what you recorded.</p>                | <p>• <b>NATURE NOTEBOOK</b><br/>Prompt for General Science in Outdoor Work Quick Link.</p>                                                                                                                                                                                                                                                                              |
| WEEK 6 | <p><b>10m Natural History: Grade 3 - Lesson 18</b><br/><i>The Crow</i></p> <p><b>Materials:</b> Christian Liberty Nature Reader 4</p> <p>PREP: Read Teacher Tip</p> <p>→ <b>INTRO</b><br/>Tell me something that is similar and something that is different about the purple martin and the kingbird. How is the quail similar to and different from them? Look at the picture on p.23. Have you seen or heard a crow before?</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/>p.23-25 "Caw! Caw!" - "are very wise."</p> <p>• Compare the crow and the purple martin. Why do you think the crow might not be as sensitive to changes in the ecosystem as the purple martin?</p> <p>→ <b>SUPPLEMENTAL</b><br/>∞ Audio &amp; Image Link: All About Birds: Crow</p> | <p>★ <b>TEACHER TIP</b><br/>It can sometimes be unnerving for students not to get an immediate answer to a question, so encourage them to sit with big questions, like the one in today's suggested discussion, while they continue to make observations.</p> <p>• <b>OUTDOOR WORK</b><br/>Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p> |

## Term 2

### WEEK 7

#### 15m General Science: Grade 3 - Lesson 19

*Dissecting an Owl Pellet Activity*

Materials: Nature Journal, owl pellet dissection kit

PREP: Read Teacher Tip

##### → INTRO

Scientists learn about creatures, like Buzztail, by studying their habitats, their food, and by doing experiments. For example, we can learn a lot about an owl and its habitat by finding out what it's eating. After an owl eats, it regurgitates a pellet that contains all of the undigestible parts of its prey. What kinds of things do you think might be in an owl pellet? Let's find out by dissecting one!

##### → ACTIVITY

Spend about 10 minutes dissecting an owl pellet. If you need a bone chart, there is one at the link. You will not finish this activity today, so take your time.

∞ Image Link: All About Owl Pellets

##### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- Tell about what you found or what you recorded.

##### ★ TEACHER TIP

Students have 2 lessons scheduled to spend on their owl pellet dissection. If there is significant interest, provide additional time in the afternoon.

##### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 7

#### 10m Natural History: Grade 3 - Lesson 19

*The Owl and Their Anatomy*

Materials: Christian Liberty Nature Reader 4

##### → INTRO

Remind me what we read about last time. Look at the picture on p.26. Do you know who this is? How does this bird look different from the others you have read about? We can't see it in this picture, but the owl has very special feathers. Some people think they are like the soft, downy feathers that lie underneath and keep birds warm, but they are not. They have something more like fringes on the edges. Let's see what else we can learn about them in the reading.

##### → READ, NARRATE, & DISCUSS

p.27-32 "Did you ever" - "from the perch."

##### → SUPPLEMENTAL

∞ Audio & Image Link: The Silent Flight of Owls Explained

∞ Audio & Image Link: All About Birds: Barred Owl

∞ Audio & Image Link: All About Birds: Great Horned Owl

##### ★ TEACHER NOTE

This author states that owls' feathers are tipped with down, but this is false according to the Audubon Society. If reading aloud, teachers can begin the third sentence of the second paragraph with "The owl has unique feathers, making their flight..." or advise students as noted in the introduction. The first link contains a photograph for those who are interested.

∞ Image Link: The Silent Flight of Owls Explained

##### ★ TEACHER NOTE

This reading has two owls linked because the author describes the call of the barred owl, but the image is of the great horned owl, whose range is more widespread.

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 8

#### 15m General Science: Grade 3 - Lesson 20

*Dissecting an Owl Pellet Activity*

Materials: Nature Journal, owl pellet dissection kit

##### → ACTIVITY

Spend about 10 minutes dissecting an owl pellet. If you don't finish today, you can come back to it during an afternoon. If you need a bone chart, you can find one at the link.

∞ Image Link: All About Owl Pellets

##### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- Tell about what you found or what you recorded.

##### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 8

#### 10m Natural History: Grade 3 - Lesson 20

*More About Owls and Their Nesting Habits*

Materials: Christian Liberty Nature Reader 4

##### → INTRO

Tell me what you recall about the owl. Today, we'll read about how they raise their young.

##### → READ, NARRATE, & DISCUSS

p.33-37 "Owls do much" - "to hunt."

- Why do you suppose that many large, predatory birds (raptors) nest in the winter? We always have to be very careful about interfering with wildlife because sometimes we can do harm without realizing it. What would you do if you found owlets, like the ones found by the men in the silo? Did you know that there are special people, called wildlife rehabilitators, who know how to

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.





## Term 2

take care of such animals?

### → SUPPLEMENTAL

∞ Video Link: Barred Owl Cam Highlights (5:00)

## WEEK 9

### 15m General Science: Grade 3 - Lesson 21

*Leaper in Context*

Materials: Buzztail and Leaper

### → INTRO

Today, we begin the story of Leaper. Look at the picture on the cover. What sort of creature is Leaper? Have you ever seen a live fish before?

### → READ, NARRATE, & DISCUSS

p.66-75 "It was late" - "three inches long."

• Leaper's egg hatches in a freshwater biome. What other plants and animals live in his biome? What do you know so far about the non-living parts of the ecosystem that affect Leaper?

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

## WEEK 9

### 10m Natural History: Grade 3 - Lesson 21

*An Owl of Your Choice*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

### → INTRO

Tell me what you recall about the owl. The book has many different kinds of owls in it, and you can read about any of them, but let's just choose one or two today.

### → READ, NARRATE, & DISCUSS

selection from p.40-59

• How does the owl's color and pattern serve it? Do you think it is more sensitive to changes in the ecosystem, like the purple martin, or more adaptable, like the crow? Why?

### → SUPPLEMENTAL

∞ Website Link: All About Birds Owl List

### ★ TEACHER TIP

Teachers may wish to select one or two owls from the variety based on those that are in their locale, or they may wish to let students select based on their interest.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## WEEK 10

### 15m General Science: Grade 3 - Lesson 22

*Leaper in the Food Chain*

Materials: Buzztail and Leaper

### → INTRO

Tell me what happened to Leaper in the last passage. Today, we'll hear a bit more about Leaper's life and the other creatures that live with him.

### → READ, NARRATE, & DISCUSS

p.77-87 "One afternoon" - "shut behind him."

• What do you know so far about the food chain here? Where is Leaper on the food chain? Draw or diagram, if helpful.

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

## WEEK 10

### 10m Natural History: Grade 3 - Lesson 22

*A Quiet Approach and Watching Hummingbirds*

Materials: Christian Liberty Nature Reader 4

### → INTRO

Have you spent time watching birds? What would you say is the most important thing to do when watching them? Have you ever had a bird come up close to you? Let's find out what our book suggests.

### → READ, NARRATE, & DISCUSS

p.61-66 "Just think about" - "see hummingbirds."

• Show me how you would want to go bird watching, so that you don't scare away the birds.

### → SUPPLEMENTAL

∞ Audio & Image Link: All About Birds: Ruby-throated Hummingbird

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 2

### WEEK 11 15m General Science: Grade 3 - Lesson 23

*Leaper's Ecosystems*

Materials: Buzztail and Leaper

#### → INTRO

Recall what you heard about Leaper in the last passage. Leaper is getting ready for a big change, so let's find out about that!

#### → READ, NARRATE, & DISCUSS

p.88-95 "A few miles" - "to the east."

• Leaper is moving from the freshwater biome into the marine biome. What do you notice about how the living and non-living parts of the ecosystem are changing around him?

### WEEK 11 10m Natural History: Grade 3 - Lesson 23

*Look for Color, Beaks, Wings, and Tails*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

#### → INTRO

What are some of your favorite birds that you learned about this term? Tell me about how the mother hummingbird teaches her young to drink nectar. Let's read today about some interesting things to notice when bird watching.

#### → READ, NARRATE, & DISCUSS

p.67-71 "When you want" - "on the ground."

• Have you noticed some of these features in birds you have seen or read about? How do you think each feature helped the bird to fit its purpose in the ecosystem?

#### ★ TEACHER TIP

Encourage students to think of some examples from their observations to connect to the text.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 12 15m General Science: Grade 3 - Lesson 24

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

Questions will come from:

Buzztail and Leaper, Nature Journal

### WEEK 12 10m Natural History: Grade 3 - Lesson 24

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

Questions will come from:

Christian Liberty Nature Reader 4

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## Term 3

### WEEK 1

#### 15m General Science: Grade 3 - Lesson 25

*Observing Seasonal Changes Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

##### → INTRO

Spend about 10 minutes observing two easy-to-see signs of the season: clouds and birds. What colors and shapes do you notice in the sky? What colors and patterns do you see in the birds? Are the clouds moving or changing? Do you think they are cirrus, stratus, or cumulus? How do the birds behave? Do you know these birds? What colors and markings do you notice? Do the clouds and birds seem different today than they did last term?

##### → OBSERVE

##### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- Tell about what you recorded.

##### ★ TEACHER TIP

There is no implied connection between clouds and birds; they are simply easy-to-notice indicators of seasonal change. If teachers prefer to do an object lesson on birds today, this could be a good day. Questions are embedded in the lesson to help, as needed. If preferred, teachers can instead refer to the Handbook of Nature Study or allow questions to flow naturally between student and teacher.

##### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 1

#### 10m Natural History: Grade 3 - Lesson 25

*Turtles*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

##### → INTRO

Turtles and snakes are cousins in the reptile family. We'll read a bit about several different members in the family, starting today with the turtles. What do you know about turtles?

##### → READ, NARRATE, & DISCUSS

p.137-139 "A turtle walks" - "walked away."

- What do you think turtles have in common with their reptile cousin, the snake? Draw or diagram the parts of the turtle discussed in the reading.

##### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

##### ★ TEACHER TIP

If students need support drawing or diagramming, the teacher might do the activity with them in tandem or act as their scribe.

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 2

#### 15m General Science: Grade 3 - Lesson 26

*Leaper in the Food Chain*

Materials: Buzztail and Leaper

PREP: Read Teacher Tip

##### → INTRO

Do you recall where we left Leaper?

##### → READ, NARRATE, & DISCUSS

p.96-105 "When spring came" - "He was caught."

- How has Leaper's food chain changed at this stage in his life? What are the reasons that Leaper's food chain changes?

##### ★ TEACHER TIP

It has been several weeks since students left Leaper, so they may need some help recalling where the story left off. It is okay to look back at the pictures on previous pages or even read the last few sentences from p.94, if they need this help.

##### ★ TEACHER NOTE

Page 105 doesn't actually have the page number, so you may need to indicate it before the reading.

### WEEK 2

#### 10m Natural History: Grade 3 - Lesson 26

*More About Turtles*

Materials: Christian Liberty Nature Reader 4

##### → INTRO

Tell me what you remember about turtles from the last passage. Today, let's find out about their young and what they eat.

##### → READ, NARRATE, & DISCUSS

p.140-141 "A farmer noticed" - "to be quite old."

- Can you diagram a food chain for the turtle? What is missing from the reading? Who do you think preys on turtles?

##### ★ TEACHER NOTE

Note that many of the turtle passages in the Nature Reader comment about keeping these creatures as pets and about whether they are used for food. In many locations, it is now illegal to trap or keep them, however.

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### WEEK 3

#### 15m General Science: Grade 3 - Lesson 27

*Leaper Moves*

Materials: Buzztail and Leaper

##### → INTRO

What happened to Leaper in the last part of the story? What do you think is going to happen next?

##### → READ, NARRATE, & DISCUSS

p.106-115 "Leaper thrashed" - "pushed on upstream."

- Did you notice the changes that humans made that helped Leaper on his migration? What were they? How did they help?

### WEEK 3

#### 10m Natural History: Grade 3 - Lesson 27

*Box and Painted Turtles*

Materials: Links

PREP: Read Teacher Tip

##### → INTRO

Tell me what you recall so far about turtles. We'll read about two different kinds of turtles today, but these turtles aren't in our book. Have you seen or heard about the box turtle or the painted turtle? Why do you think they are called those names?

##### → READ & NARRATE

∞ Article Link: Museum of Discovery: Ornate Box Turtle

##### → READ, NARRATE, & DISCUSS

∞ Article Link: Wildlife Medical Clinic: Let's Paint a Story

- How are these two turtles similar and how are they different? How do you think their special features help them to fit their purpose in the ecosystem?

##### → SUPPLEMENTAL

∞ Video Link: Eastern Box Turtle (1:49)

∞ Video Link: Western Painted Turtle (1:19)

##### ★ TEACHER TIP

Some of the lessons on reptiles require supplementation. In this lesson, read and narrate the two portions separately because they are on different, though related topics. Then discuss.

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 4

#### 15m General Science: Grade 3 - Lesson 28

*Life Cycle of a Fish*

Materials: Buzztail and Leaper

##### → INTRO

Remind me what happened to Leaper last time. Do you think he will finish the journey? It seems like a hard journey. Why is he going all this way?

##### → READ, NARRATE, & DISCUSS

p.116-128 "On they went" - "another time."

- How do different parts of the ecosystem affect Leaper's life cycle? This could include the land, seasonal changes, special features in his body, other plants and animals, etc. Draw or diagram, if helpful.

### WEEK 4

#### 10m Natural History: Grade 3 - Lesson 28

*Wood and Snapping Turtles*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

##### → INTRO

Who were the two turtles we read about last time? We'll read about two more turtles today: the wood turtle and the alligator snapping turtle. Can you tell from their names what their habitats might be?

##### → READ & NARRATE

p.142-143 "The wood turtle" - "vegetables and berries."

##### → READ, NARRATE, & DISCUSS

∞ Article Link: NC Zoo: Oh Snap!

- How are these turtles similar and different? How do you think their special features help them to fit their purpose in the ecosystem?

##### ★ TEACHER TIP

Again in this lesson, read and narrate the two portions separately because they are on different, though related topics. Then discuss.

##### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### → SUPPLEMENTAL

- ∞ Video Link: Wood Turtle (1:28)
- ∞ Video Link: Snapping Turtle (0:33)

### WEEK 5

#### 15m General Science: Grade 3 - Lesson 29

*Observing Trees Activity*

Materials: Nature Journal

### → INTRO

Spend about 10 minutes comparing the buds of two trees. If they have already bloomed, then compare the young growth at the ends of their twigs.

### → OBSERVE

### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

### WEEK 5

#### 10m Natural History: Grade 3 - Lesson 29

*Snakes*

Materials: Christian Liberty Nature Reader 4

### → INTRO

Tell me about the last two turtles that we read about. Today, let's read about their reptile cousins, the snakes. What do you recall about snakes? Note that this author uses the term "poisonous" when the correct term is "venomous." A poison is something that is ingested; a venom is something that is injected.

### → READ, NARRATE, & DISCUSS

p.153-155 "If you are" - "the eggs whole."

- Draw or diagram the parts of the snake discussed in the reading. What makes a snake a reptile and not a mammal or a bird? What makes a snake a snake and not a turtle?

### → SUPPLEMENTAL

- ∞ Image Link: Growing with Science: Snake Shed
- ∞ Video Link: Snake Up Close (1:54)

### ★ TEACHER NOTE

Note that this author uses the term "poisonous" when the correct term is "venomous." A poison is something that is ingested; a venom is something that is injected.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 6

#### 15m General Science: Grade 3 - Lesson 30

*Observing Seasonal Changes Activity*

Materials: Nature Journal

PREP: Read Teacher Tip

### → INTRO

Spend about 10 minutes observing two easy-to-see signs of the season: clouds and birds. What colors and shapes do you notice in the sky? What colors and patterns do you see in the birds? Are the clouds moving or changing? Do you think they are cirrus, stratus, or cumulus? How do the birds behave? Do you know these birds? Do the clouds and birds seem different today than they did last term?

### → OBSERVE

### → NARRATE & DISCUSS

Use a nature journal to record what the student noticed.

- Tell about what you recorded.

### ★ TEACHER TIP

This could be a good opportunity for an object lesson with birds or clouds.

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 6

#### 10m Natural History: Grade 3 - Lesson 30

*More About Snakes*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

### → INTRO

Remind me what we read about in the last passage. We'll learn a bit more about snakes today.

### → READ, NARRATE, & DISCUSS

p.156-158 "A poisonous snake" - "destroy many snakes."

- Draw or diagram a food chain for the snake, based on what you read. Does it intersect with other food chains that you know about? You could add them to this food chain and make it a food web!

### → SUPPLEMENTAL

- ∞ Image Link: Growing with Science: Snake with Full Belly

### ★ TEACHER TIP

Supplemental links for optional support are provided at the END of the lesson. They can be used at any point to help generate interest during the introduction, to enliven the lesson itself, or to add to discussion later. They are NOT required and should NOT take away from the narration and discussion. Experiment and see what works best for your student(s).

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### WEEK 7

#### 15m General Science: Grade 3 - Lesson 31

*Lots of Animals Move!*

Materials: Going Home

##### → INTRO

What was Leaper's migration like? Do you know about how far or how long the migration was? Do you know of any other creatures that migrate? This book is about the annual migrations that many animals make each year. One of our migrating animals is on the cover. Do you know who he is? Do you know any other migrating friends that you might find in our book?

##### → READ, NARRATE, & DISCUSS

Going Home

- Imagine yourself as one of the animals in the book. What would it take to travel on such a long journey? What obstacles might you face? How would you find your way?

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 7

#### 10m Natural History: Grade 3 - Lesson 31

*Venomous Snakes*

Materials: Christian Liberty Nature Reader 4

PREP: Read Teacher Tip

##### → INTRO

Some people think that snakes are scary. Why do you think that is? Tell me what you have learned about how snakes move along and about the venomous fangs that some have. Let's read about three kinds of venomous snakes today. In addition to rattlesnakes, these are all of the venomous snakes in North America. You should learn how to identify the ones near you at your local nature preserve or state park.

##### → READ, NARRATE, & DISCUSS

p.162-167 "The copperhead" - "and birds' eggs."

- What do you do if you encounter a snake that you think might be venomous? What features could you look for to help you identify the snake without getting too close?

##### → SUPPLEMENTAL

∞ Video Link: Rattlesnake Up Close (1:04)

#### ★ TEACHER NOTE

It is best for students to learn to identify their local venomous snakes from a local nature preserve or state park.

#### ★ TEACHER TIP

If students are not familiar with a "dumbbell" shape, then suggest a different image, such as an hourglass or Hershey's Kisses.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 8

#### 15m General Science: Grade 3 - Lesson 32

*Mapping the Arctic Migration*

Materials: Going Home, globe

##### → INTRO

Tell me about some of the creatures that we read about last time. Today, we'll follow some of the journeys on our globe.

##### → ACTIVITY

Going Home

Notice the map toward the end of the book and find this view on your globe. Find where you are on the globe (teacher's help, as needed). Using the time you have, choose an animal, reread their poem, and trace their migration on the globe.

##### → DISCUSS

Which trip was surprising to you? Most interesting? Who do you think had the longest journey?

#### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 8

#### 10m Natural History: Grade 3 - Lesson 32

*Water Snakes*

Materials: Christian Liberty Nature Reader 4, article link

PREP: Read Teacher Tip

##### → INTRO

Tell me what you learned about venomous snakes. Let's take a closer look at some of North America's nonvenomous snakes, beginning with the water snake. While many different kinds of snakes like to swim, the water snake is a specific kind of snake. They are only native in the east, but are also found in the west, and so could really be found anywhere.

##### → READ

p.169 "The water snake" - "by a bass."

##### → READ, NARRATE, & DISCUSS

∞ Article Link: Museum of Life and Science: Northern Water Snake

- How do you think the water would affect a snake's movement? How does it affect your movement when you swim or play in the water?

#### ★ TEACHER TIP

Some of the lessons on reptiles require supplementation. In this lesson, read the very short passage from the text, followed by the article. Then narrate and discuss as a single topic.

#### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.



## Term 3

### → SUPPLEMENTAL

∞ Video Link: Common Water Snake (2:39)

### WEEK 9

#### 15m General Science: Grade 3 - Lesson 33

*Museum Curator Activity*

Materials: Nature Journal, found nature items, collecting containers or other means of collecting

PREP: Read Teacher Tip

### → INTRO

Over the next few weeks, you will curate your own ecosystem museum. Your museum may contain as many or as few Things as you want to represent your ecosystem. Your museum can be in whatever form(s) you like and is most appropriate for the Things you want to represent. It could be a collection of samples, photographs, drawings, or any combination of those. An example is shown at the link. Spend about 10 minutes today deciding what ecosystem you want to tell about through your museum. Looking through your nature journal and any collected nature items will help.

∞ Image Link: Museum Example

### → ACTIVITY

### → NARRATE & DISCUSS

Share what was decided, what ideas the student has, and whether any help is needed.

### ★ TEACHER TIP

If there is significant interest, provide additional time in the afternoons.

### • NATURE NOTEBOOK

Prompt for General Science in Outdoor Work Quick Link.

### WEEK 9

#### 10m Natural History: Grade 3 - Lesson 33

*King Snakes*

Materials: Christian Liberty Nature Reader 4, article link

PREP: Read Teacher Tip

### → INTRO

What do you remember about water snakes? Let's read a little about king snakes today, of which there are many different cousins in the family all over North America. Why do you think they are called 'king'?

### → READ

p.170-171 "The king snake" - "bold and fearless."

∞ Image Link: Museum of Life and Science: Scales

### → READ, NARRATE, & DISCUSS

∞ Article Link: National Park Service: Sierra Mountain Kingsnake

• This snake can take on many different colors and patterns, so what makes a king snake a king snake?

### → SUPPLEMENTAL

∞ Image Link: National Park Service: The King of Snakes

### ★ TEACHER TIP

Again in this lesson, read the short passage from the text, followed by the article. Then narrate and discuss as a single topic. The link provided with the short passage in the reader is intended to help with the mention of smooth scales versus rigid scales. If this is not of interest, then the image is not necessary.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

### WEEK 10

#### 15m General Science: Grade 3 - Lesson 34

*Museum Curator Activity*

Materials: Nature Journal, found nature items, collecting containers or other means of collecting

### → ACTIVITY

Continue working on your museum by collecting additional Things or organizing those already collected.

### → NARRATE & DISCUSS

Share what was completed, what ideas the student has, and whether any help is needed.

### WEEK 10

#### 10m Natural History: Grade 3 - Lesson 34

*Garter Snakes*

Materials: Christian Liberty Nature Reader 4, article link

PREP: Read Teacher Tip

### → INTRO

Tell me what we read about last time. One of the most common types of snakes in North America is the garter snake. This is another one that can be found in many different variations all over the continent and is one of the most commonly seen snakes.

### → READ

p.173 "The garter snake" - "on their own."

### ★ TEACHER TIP

Again in this lesson, read the short passage from the text, followed by the article. Then narrate and discuss as a single topic.

### • OUTDOOR WORK

Weekly walks and daily outdoor time are vital! Resources in Quick Links.

## Term 3

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|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                | <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/> ∞ Article Link: Ames Free Library: Common Garter Snake (from "Behind Queset Garden's reflecting pool")</p> <p>• Snakes can sometimes appear to have striking colors and patterns, but then they seem to disappear in their natural habitat. What do you imagine a garter snake might appear to be in a forest or pond habitat?</p> <p>→ <b>SUPPLEMENTAL</b><br/> ∞ Image Link: Audubon Portland: Patient of the Week</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                     |
| <b>WEEK 11</b> | <p><b>15m General Science: Grade 3 - Lesson 35</b><br/> <i>Museum Curator Activity</i></p> <div data-bbox="300 552 1057 600"> <p>Materials: Nature Journal, found nature items, collecting containers or other means of collecting</p> </div> <p>→ <b>ACTIVITY</b><br/> Finish your museum by organizing and arranging the last of your Things.</p> <p>→ <b>NARRATE &amp; DISCUSS</b><br/> Now you are the docent! Tell about your ecosystem museum.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                     |
| <b>WEEK 11</b> | <p><b>10m Natural History: Grade 3 - Lesson 35</b><br/> <i>Bull Snakes</i></p> <div data-bbox="300 816 1057 848"> <p>Materials: article link</p> </div> <p>→ <b>INTRO</b><br/> Remind me about garter snakes - what do you know about them? The last snake we'll learn about this term is the bull snake. It is an interesting snake that is more common in the western part of the continent. Its eastern cousin, called the pine snake, is important because it is endangered and many people don't know about it.</p> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b><br/> ∞ Article Link: Bull Snake Encounter</p> <p>• Many creatures use mimicry as a defense, like the bull snake mimicking a rattlesnake. Can you think of other creatures that you have learned about that mimic someone else in their biome?</p> <p>→ <b>SUPPLEMENTAL</b><br/> ∞ Image Link: Smithsonian National Zoo: Northern Pine Snake<br/> ∞ Image Link: Museum of Natural Science: Pine Snake Nests</p> | <p>• <b>OUTDOOR WORK</b><br/> Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p>                                                                                                                                                          |
| <b>WEEK 12</b> | <p><b>15m General Science: Grade 3 - Lesson 36</b><br/> <i>Exam Week</i></p> <p>→ <b>EXAMS</b><br/> Answer question(s) related to course.</p> <p>Questions will come from:<br/> Going Home, Buzztail and Leaper, Nature Journal</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                     |
| <b>WEEK 12</b> | <p><b>10m Natural History: Grade 3 - Lesson 36</b><br/> <i>Exam Week</i></p> <p>→ <b>EXAMS</b><br/> Answer question(s) related to course.</p> <p>Questions will come from:<br/> Christian Liberty Nature Reader 4</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>★ <b>TEACHER NOTE</b><br/> Students may use other readings in the book not used in these lesson plans for extra reading practice, as appropriate!</p> <p>• <b>OUTDOOR WORK</b><br/> Weekly walks and daily outdoor time are vital! Resources in Quick Links.</p> |



# Science: Grade 3

## Examination

### Term 1

#### **General Science: Grade 3**

- Tell about or diagram the ways that water affects the living things in the ecosystem.

#### **Natural History: Grade 3**

- Christian Liberty Nature Reader 4: Draw or diagram one of the birds you saw or read about this term, showing how its body serves it in the ecosystem.

### Term 2

#### **General Science: Grade 3**

- Tell about or diagram to explain a life cycle or food chain.

#### **Natural History: Grade 3**

- Christian Liberty Nature Reader 4: Draw or diagram one of the birds you saw or read about this term, showing how its body serves it in the ecosystem.

### Term 3

#### **General Science: Grade 3**

- Tell about or draw reasons that animals migrate.

#### **Natural History: Grade 3**

- Christian Liberty Nature Reader 4: Draw or diagram one of the reptiles you saw or read about this term, showing how its body serves it in the ecosystem.